

# CYPHER

## Training School on Scientific Machine Learning for Digital Twins

*System Identification and machine  
learning for predicting reacting dynamics  
Hands-on session*

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# Short instructions

Scripts and data available here:

[https://github.com/ndoan-icl/CYPHER\\_MLSchool\\_Brussels2026](https://github.com/ndoan-icl/CYPHER_MLSchool_Brussels2026)

Search for ndoan-icl on google colab

Slides also available on the git repo

Alternatively, after copying the git repo:

Create an environment with:

```
conda create --name cypher --file requirements.txt
```

# Short instructions

- Alternatively, install the following libraries (beyond the usual ones – numpy, scipy, matplotlib):
  - tensorflow
  - impulseest
  - scikit-learn
  - scikit-opt
  - jupyterlab

# Structure of the session

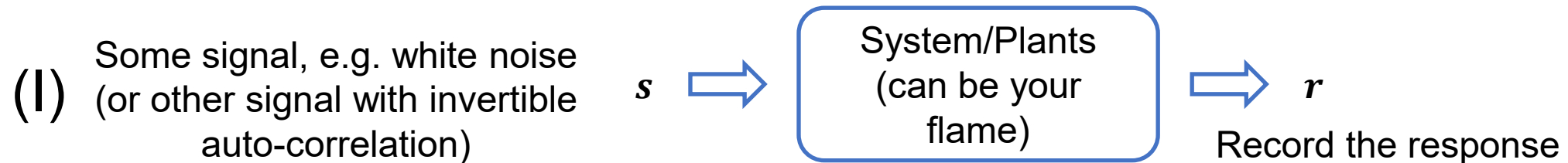
1. **5-minute recap**
2. Guide through example notebook: System Identification
3. Guide through example notebook: Lorenz system
4. Description of the exercise: Learning flame dynamics

# Very short recap on System Identification

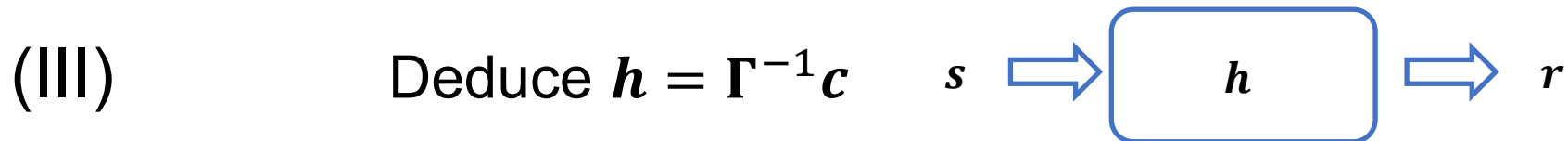
$$\rightarrow \mathbf{h} = \mathbf{\Gamma}^{-1} \mathbf{c} \text{ or } \mathbf{h}_{\alpha} = (\mathbf{\Gamma}^T \mathbf{\Gamma} + \alpha^2 \mathbf{R} \mathbf{R}^T)^{-1} \mathbf{\Gamma}^T \mathbf{c}$$

- If  $s$  is white noise,  $\mathbf{\Gamma} = \mathbf{I}$  (identity matrix)

- Summary



(II) Compute  $\mathbf{\Gamma}, \mathbf{c}$



(IV) With  $\mathbf{h}$ , estimate  $\mathbf{r}$  for any  $s$

# Modelling of time series with feedforward neural networks

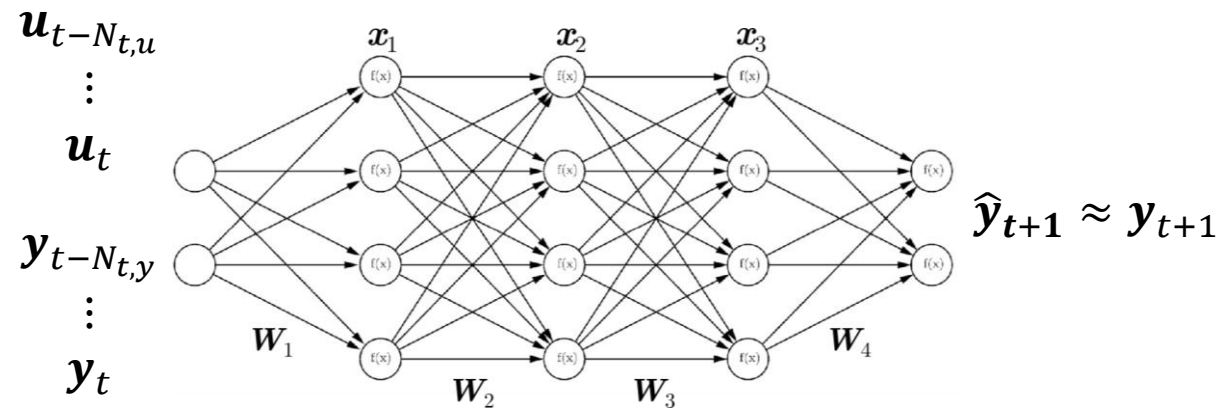
- Assume we want to model a time-dependent process:

$$\dot{\mathbf{y}}(t) = \mathcal{F}(\mathbf{y}, \mathbf{u})$$
$$\rightarrow \mathbf{y}_{t+1} = \mathcal{F}_d \left( \mathbf{y}_t, \dots, \mathbf{y}_{t-N_{t,y}}, \mathbf{u}_t, \dots, \mathbf{u}_{t-N_{t,u}} \right)$$

## Typical feedforward-based architecture

Time series of past input  
and prediction

$N_{t,u}, N_{t,y}$ : duration of  
“relevant past  
information”



# Modelling of time series with recurrent neural network

- Assume we want to model a time-dependent process:

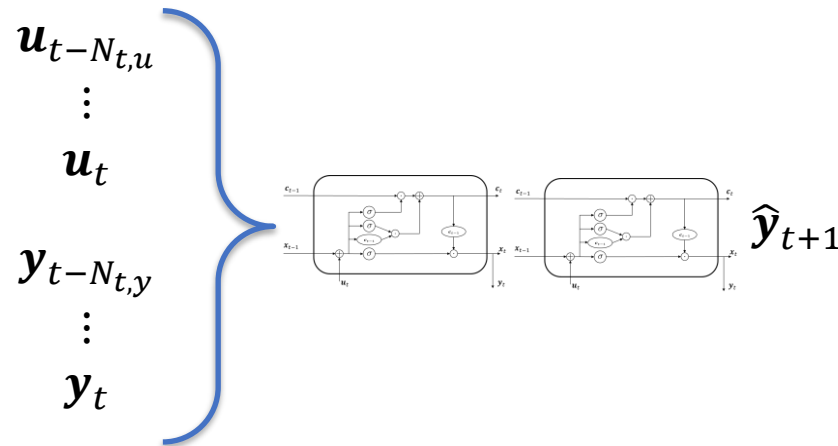
$$\dot{\mathbf{y}}(t) = \mathcal{F}(\mathbf{y}, \mathbf{u})$$

$$\rightarrow \mathbf{y}_{t+1} = \mathcal{F}_d \left( \mathbf{y}_t, \dots, \mathbf{y}_{t-N_{t,y}}, \mathbf{u}_t, \dots, \mathbf{u}_{t-N_{t,u}} \right)$$

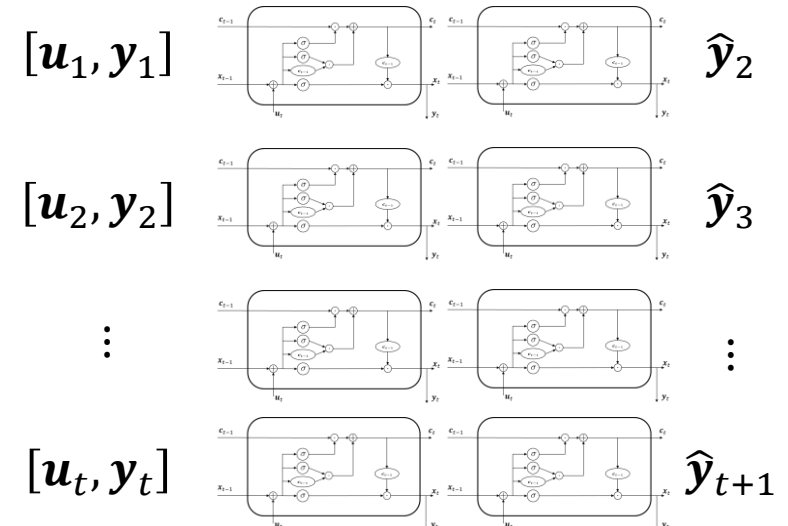
Two possible recurrent approaches with RNNs:

“Many to one” approach:

Time series of  
past input and  
prediction



“Many to many” approach:



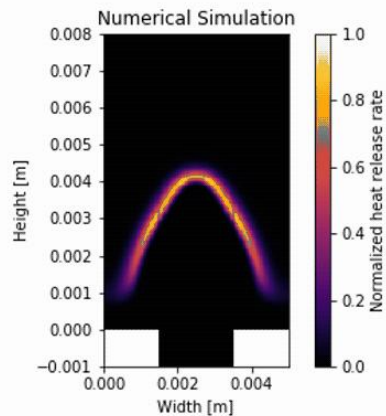
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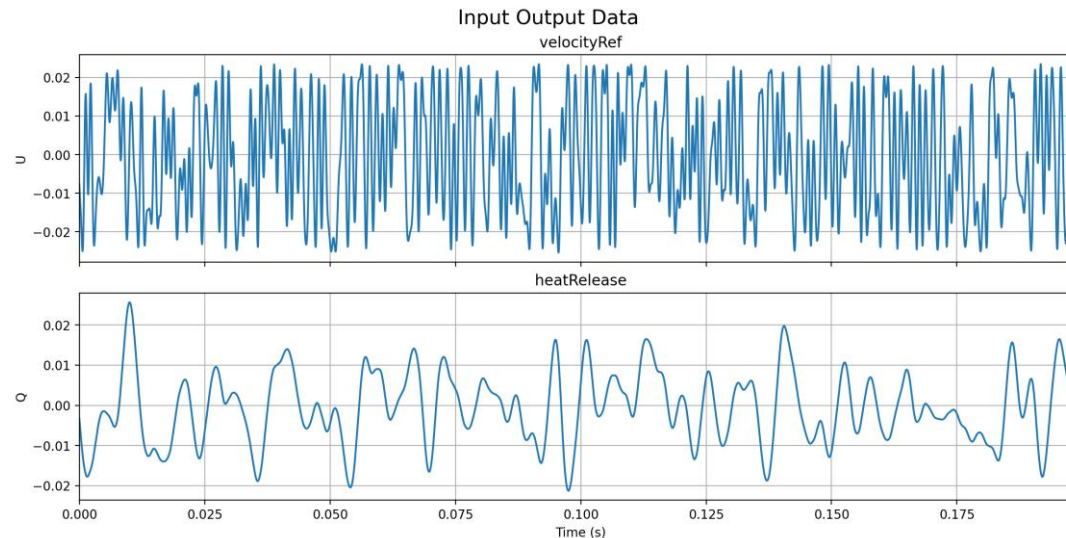


# (Linear) Flame dynamics of a laminar slit flame

- Premixed methane-air laminar slit burner (equivalence ratio of 0.8)
- Fully resolved with DNS



- Time series of input (velocity fluctuation) and output (heat release rate fluctuation) provided



- Application of System Identification to obtain flame frequency response

# Structure of the session

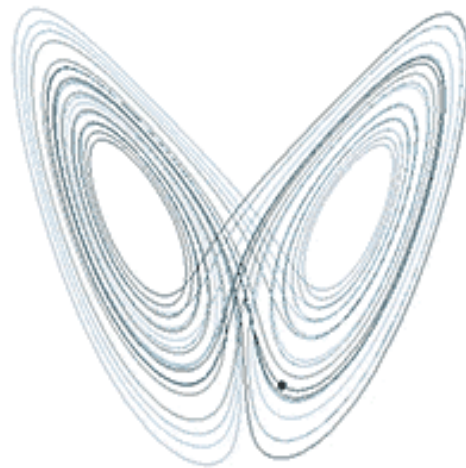
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# Guided exercise: Lorenz system

- Developed originally as a model of atmospheric convection

$$\frac{dx}{dt} = \sigma(y - x), \frac{dy}{dt} = x(\rho - z) - y, \frac{dz}{dt} = xy - \beta z$$

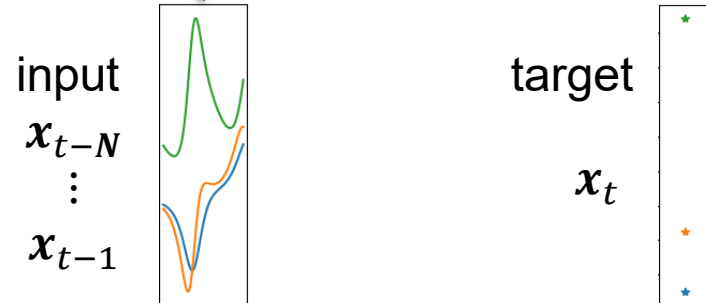
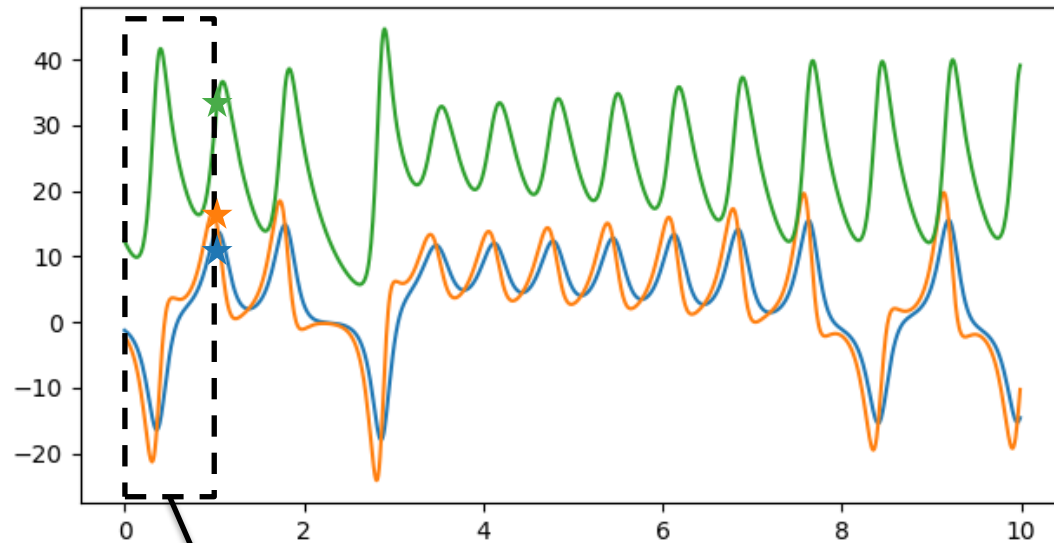
- Simplified model that exhibits chaotic behaviour.



$$\rho = 28, \sigma = 10, \beta = 8/3$$

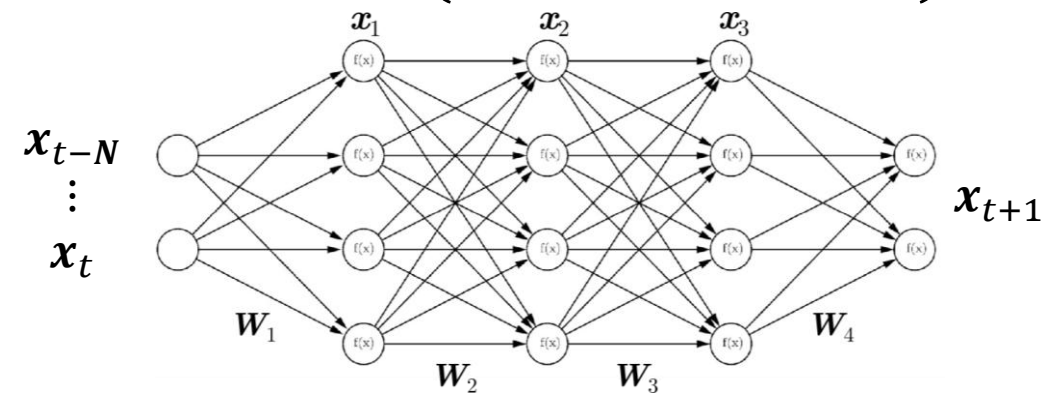
# Guided exercise: Lorenz system

Data: Time series of the Lorenz system



Objective: predict next time step based on previous ones:

$$x(t+1) = \mathcal{F}(x(t), \dots, x(t-N))$$



Steps:

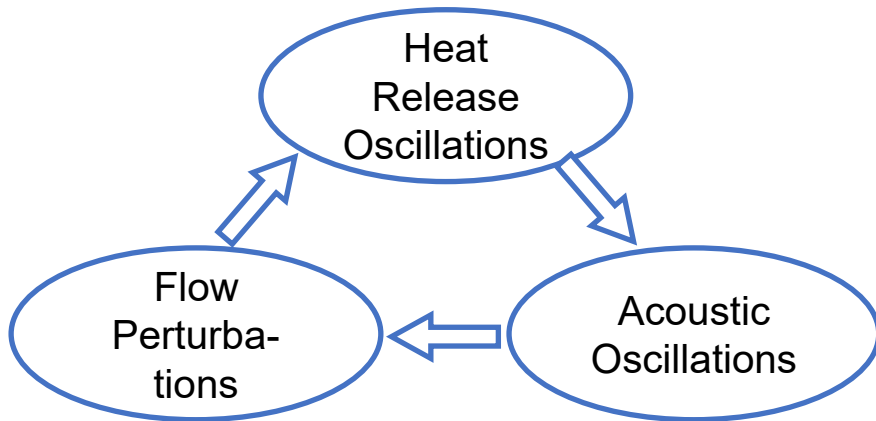
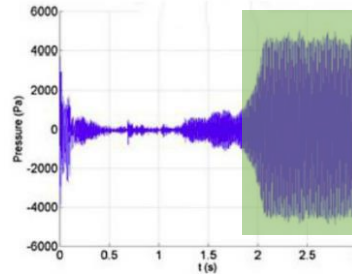
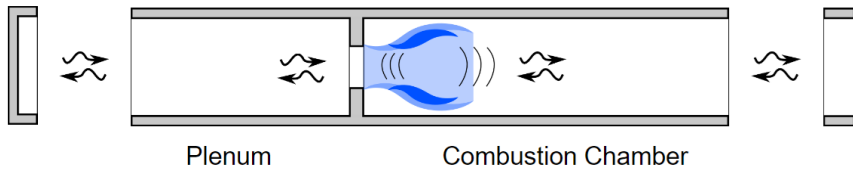
1. Restructure datasets into appropriate (input,pair)
2. Design the network
3. Train

# Structure of the session

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# Thermoacoustic instabilities result from heat release rate oscillations

- Thermoacoustic instabilities: High amplitude pressure oscillations in 'lean premixed' combustors



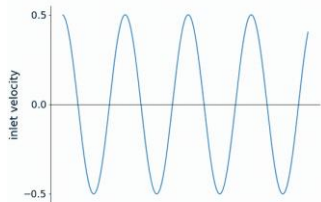
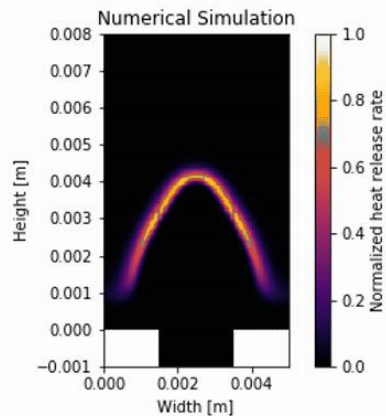
- Prediction requires a coupling between:

- *Acoustic model*

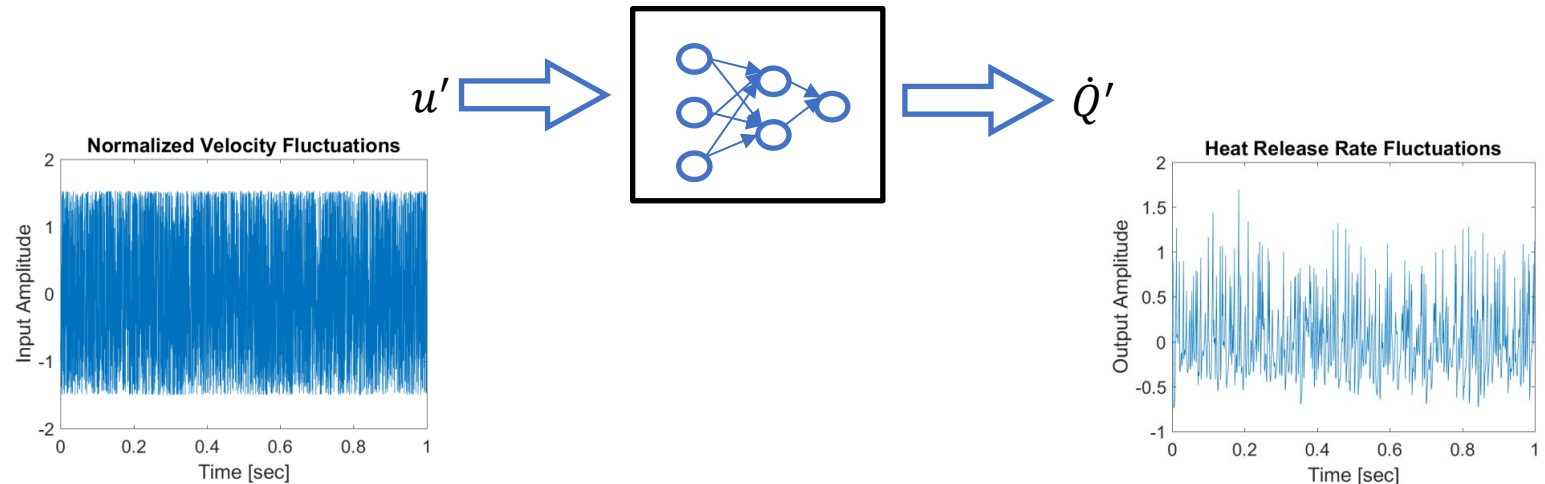
- *Flame model:  $\dot{Q}' = f(u')$*

# NN must be trained to learn flame dynamics ( $\dot{Q}' = f(u')$ ) of a laminar slit flame

- Premixed methane-air laminar slit burner (equivalence ratio of 0.8)
- Fully resolved with DNS



Objective: Design a neural networks trained using the data from broadband simulation



Similar steps as for Lorenz case except input and output are different signals

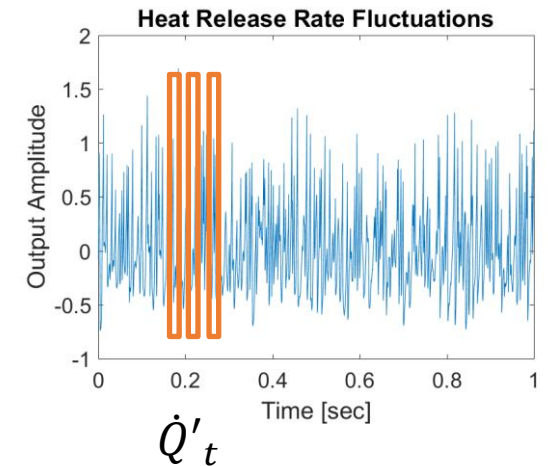
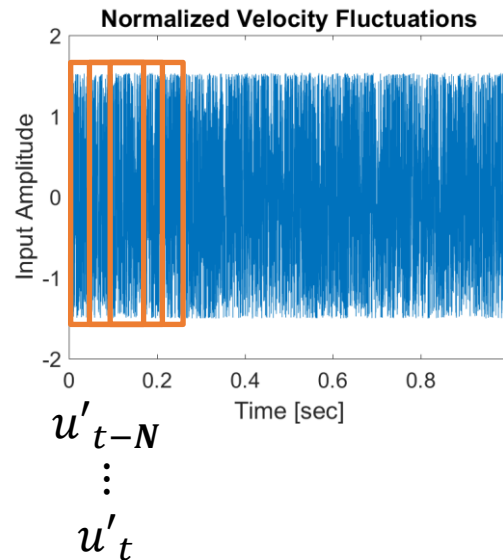
# Overall exercise

- Objective: Develop a neural network model of the flame response
- 3 training datasets provided for different amplitudes of excitations
- Two approaches possible
  - Feedforward NN
  - Recurrent neural network
- Study of the accuracy of the trained NN depending on the training dataset used
- Study on the impact of the length of the dataset used for training



# Steps for the exercise

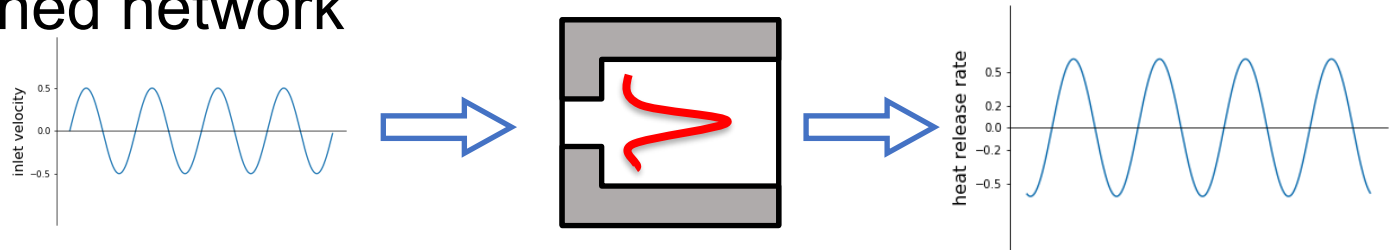
1. Preprocess the dataset
  - Consider undersampling
2. From the data provided, create the appropriate “input/output” pairs
  - Pay attention to your choice of “history” ( $N$ )
3. Design and train the network



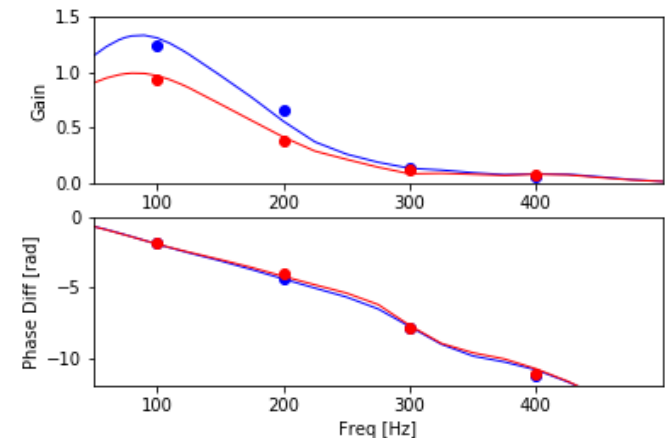
# Assessment of the trained neural network : comparison with harmonic forcing

4. Compute the flame describing function  $F(\omega, |u'|) = \frac{\dot{Q}'(\omega, |u'|)/\bar{\dot{Q}}}{u'(\omega, |u'|)/\bar{u}}$

- Create harmonic signals at given frequencies
- Feed them to your trained network

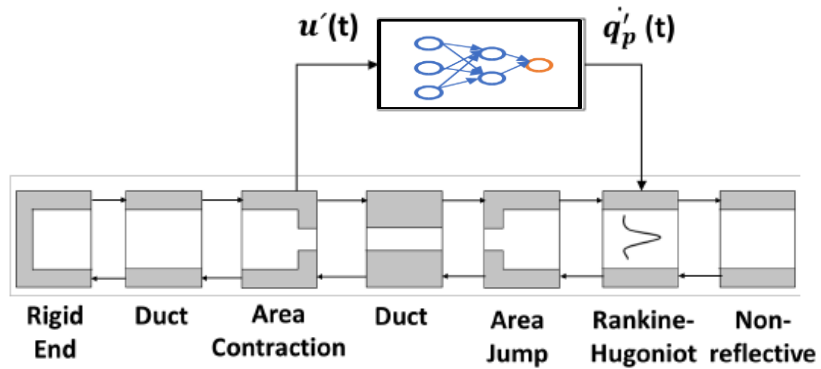


- Deduce the gain (“ratio” between input and output) and phase (normalised time shift between input/output)



# Integrated NN and acoustic network predicts accurately the thermoacoustic instability

- Trained NN with acoustic model describes the gas turbine



Stability analysis for different designs possible

